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Design Report for Silicon Photonics Edx Course

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***Abstract*—As part of the UBCx Edx course, "Silicon Photonics Design, Fabrication and Data Analysis", the students are designing, manufacturing and analyzing a Mach-Zehnder interferometer. This document presents the design proposal of one of the students.**

***Index Terms*—Silicon photonics, Mach-Zehnder interferometer**

I. INTRODUCTION

As part of the UBCx Edx course "Silicon Photonics Design, Fabrication and Data Analysis" the following design is presented of a Mach Zehnder interferometer. The goal of the design is to fabricate and measure several types of MZI circuits each with changed length difference and observe the change in free spectral range. The first MZI type uses y-splitter at one side and directional couple on the other side, creating two complementary outputs. The second MZI type uses y-splitter at both it's input and output. Loop back circuit and y branch only circuit are also implemented for calibration.

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II. THEORY

As explained in [2],[6] and repeated for completeness in the Appendix, the interferometer transfer function for both MZI described above is

$$\frac{I_o}{I_i} = \frac{1}{2}(1 + \cos(\beta \Delta L)) \tag{1}$$

The free space spectral range (FSR) which is the spacing between adjacent peaks can be calculated to be

$$FSR = \frac{\lambda^2}{\Delta L n_g} \tag{2}$$

Thus, by measuring the MZI FSR it is possible to extract the waveguide group index.

The group index can be found by forming compact model for the waveguide effective index dispassion. Using Lumerical Interconnect a frequency sweep simulation can be performed and the resulted effective index as function of wavelength can be exported to MATLAB. In the course [6], a MATLAB script is provided to find the Taylor expansion polynomial

$$n_{eff}(\lambda) \approx n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \tag{3}$$

The Taylor coefficients can be related to the physical parameters effective index, group index, and dispersion as described in the course notes [6]

$$\begin{gathered} n_{eff} = n_1 \\ n_g = n_1 - n_2\lambda_0 \\ D = -\frac{2\lambda_0 n_3}{c} \end{gathered} \tag{4}$$

The design files were submitted as part of the SiEPIC openEBL-2026-05 fabrication review process using the public GitHub repository.

III. MODELING AND SIMULATIONS

A. Waveguide Model

The MZI is implemented using strip waveguide, TE polarized with width of 0.5um and height of 0.22um. The first TE mode profile and dispersion were calculated using Lumerical mode and are plotted next

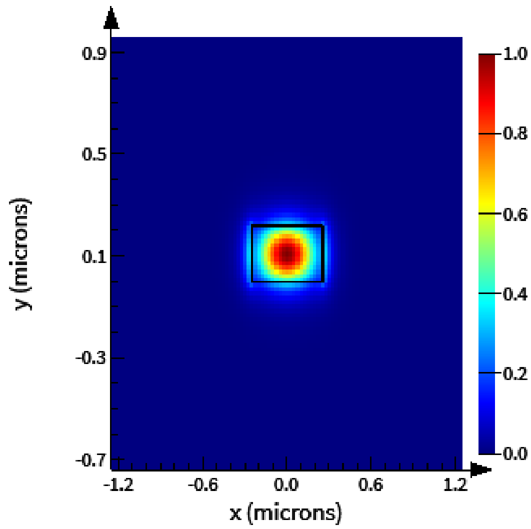

Fig. 1. Mode Profile.

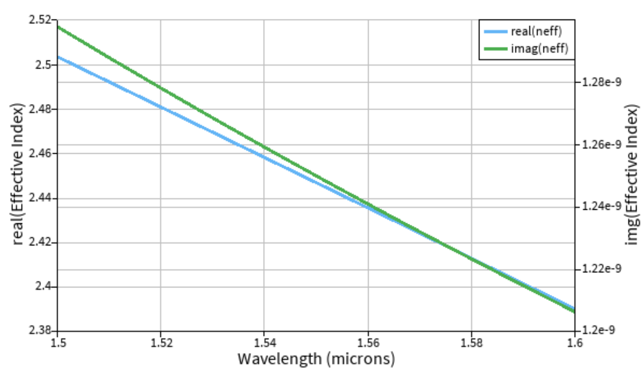

Fig. 2. Effective index versus wavelength.

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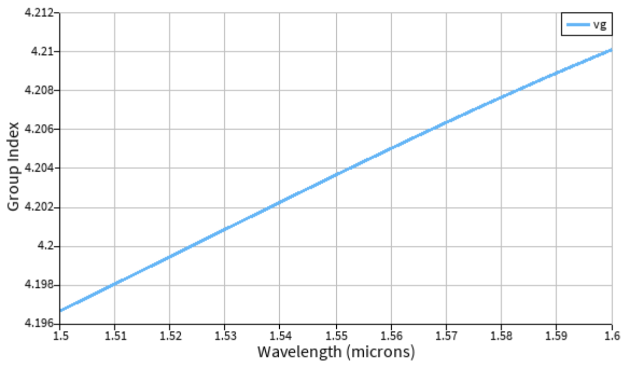

Fig. 3. Group index versus wavelength

The effective was exported to MATLAB. A MATLAB script provided by the course was used to fit a compact model for the waveguide effective index with the following result

$$n_{eff}(\lambda) \approx 2.45 - 1.13(\lambda - 1.55) - 0.049(\lambda - 1.55)^2 \quad (5)$$

From the above compact model and using (4) the group index can be calculated to be

$$n_g = 2.45 + 1.13 * 1.55 = 4.2$$

Which is similar to the group index simulated with mode as shown in figure (3).

B. Layout and MZI design variations

There are total of seven MZIs implement in the layout. MZI1 and MZI2 are implemented using y-splitter at the input and wideband directional coupler at the output. The path difference is 100um and 150um. MZI3 to MZI7 are implemented with y-splitter at both input and output and path differences of 50um, 75um, 100um, 150um and 200um. In addition, a loop back is implemented to measure the grading coupler response and an circuit to measure the response of the y splitter used.

TABLE I
LAYOUT VARIATIONS

MZI #	path difference [um]	FSR [nm]	MZI Type
1	100	5.7	Y-splitter plus coupler
2	150	3.8	Y-splitter plus coupler
3	50	11.4	y-splitter
4	75	7.6	y-splitter
5	100	5.7	y-splitter
6	150	3.8	y-splitter
7	200	2.9	y-splitter

Below is the layout as plotted in Klayout

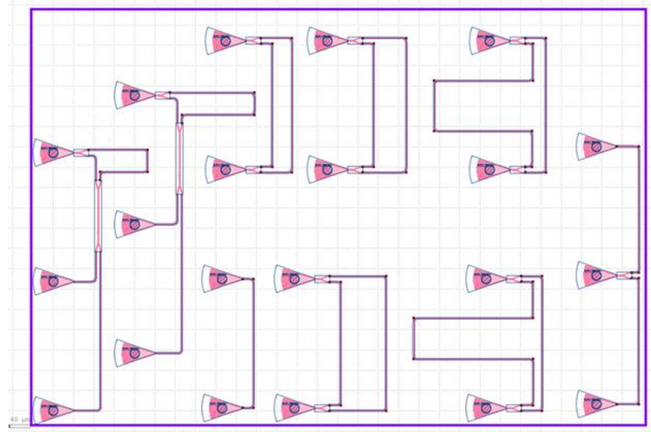

Fig. 4. Layout

C. Transmission spectrum simulation

Transmission spectrum of MZI5, which have path difference of 100um was simulated with Lumerical interconnect with below results

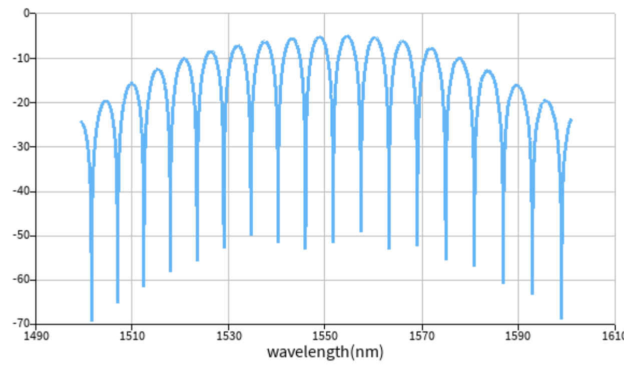

Fig. 5. MZI5 simulated transmission spectrum

From the simulated spectrum the FSR is seen to be 5.8nm, similar to the expected calculated value of 5.7nm.

IV. FABRICATION

The MZI manufacturing variability in studied next using Corner analysis using the following process. For the strip waveguide described in section III.A (width 500nm, height 220nm), it is assumed manufacturing variability in height from 215.3 to 223.1 nm and in width of 470 and 510 nm. The waveguide effective index over wavelength from 1.5um to 1.6um is calculated for the 4 corner combinations of width and height and for the nominal value using Lumerical Mode. Data is exported to MATLAB and a compact model is extracted. For each of the 5 options, effective index and, group index are calculated using (4). The results are summarized in table II.

TABLE II
CORNER ANALYSIS FOR WAVEGUIDE

h[n]	w[nm]	neff	ng
220	500	2.4468	4.2036
215.3	470	2.3724	4.2529
223.1	470	2.4039	4.2621

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215.3	510	2.4416	4.1796
223.1	510	2.4722	4.1878

Next for the MZIs implemented, with path differences of 50, 75, 100, 150 and 200um the FSR is calculated for each of the 5 options in table II. Results are summarized at table III.

TABLE III
CORNER ANALYSIS FOR MZIS

h	w	FSR1 [nm]	FSR2 [nm]	FSR3 [nm]	FSR4 [nm]	FSR5 [nm]
220	500	11.43	7.62	5.72	3.81	2.86
215.3	470	11.30	7.53	5.65	3.77	2.82
223.1	470	11.27	7.52	5.64	3.76	2.82
215.3	510	11.50	7.66	5.75	3.83	2.87
223.1	510	11.47	7.65	5.74	3.82	2.87

Finally, the transmission spectrum for MZI with 100um path difference is calculated using (1) for each of the five options as can be seen at figure 6.

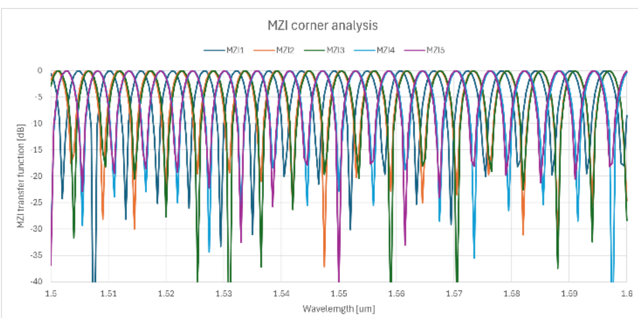

Fig. 6. Transmission spectrum corner analysis.

The devices were fabricated using 100 keV Electron Beam Lithography [1]. The fabrication used silicon-on-insulator wafer with 220 nm thick silicon on 3 µm thick silicon dioxide. The substrates were 25 mm squares diced from 150 mm wafers. After a solvent rinse and hot-plate dehydration bake, hydrogen silsesquioxane resist (HSQ, Dow-Corning XP-1541-006) was spin-coated at 4000 rpm, then hotplate baked at 80 °C for 4 minutes. Electron beam lithography was performed using a JEOL JBX-6300FS system operated at 100 keV energy, 8 nA beam current, and 500 µm exposure field size. The machine grid used for shape placement was 1 nm, while the beam stepping grid, the spacing between dwell points during the shape writing, was 6 nm. An exposure dose of 2800 µC/cm2 was used. The resist was developed by immersion in 25% tetramethylammonium hydroxide for 4 minutes, followed by a flowing deionized water rinse for 60 s, an isopropanol rinse for 10 s, and then blown dry with nitrogen. The silicon was removed from unexposed areas using inductively coupled plasma etching in an Oxford Plasmalab System 100, with a chlorine gas flow of 20 sccm, pressure of 12 mT, ICP power of 800 W, bias power of 40 W, and a platen temperature of 20 °C, resulting in a bias voltage of 185 V. During etching, chips were mounted on a 100 mm silicon carrier wafer using perfluoropolyether vacuum oil. Cladding oxide was deposited using plasma enhanced chemical vapor deposition (PECVD) in an Oxford Plasmalab System 100 with a silane (SiH4) flow of 13.0 sccm, nitrous oxide (N2O) flow of 1000.0 sccm, high-purity nitrogen (N2) flow of 500.0 sccm, pressure at 1400mT, high-frequency RF power of 120W, and a platen temperature of 350C. During deposition, chips rest directly on a silicon carrier wafer and are buffered by silicon pieces on all sides to aid uniformity.

V. MEASUREMENTS

To characterize the devices, a custom-built automated test setup [2, 6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [4]. A 90º rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [5].

Out of the seven MZI devices the result of four selected one, to simplify the report. In the next four figures the measured spectrum of MZI4, MZI5, MZI6 and MZI7 are presented

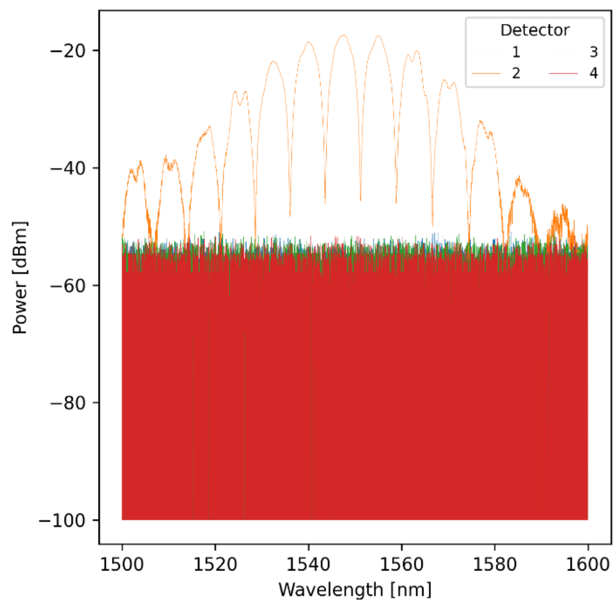

Fig. 7. Measured spectrum of MZI4, DL=75um.

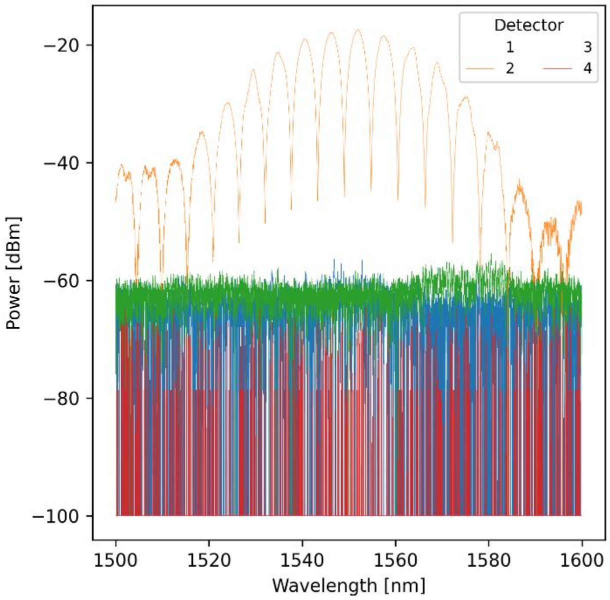

Fig. 8. Measured spectrum of MZI5, DL=100um.

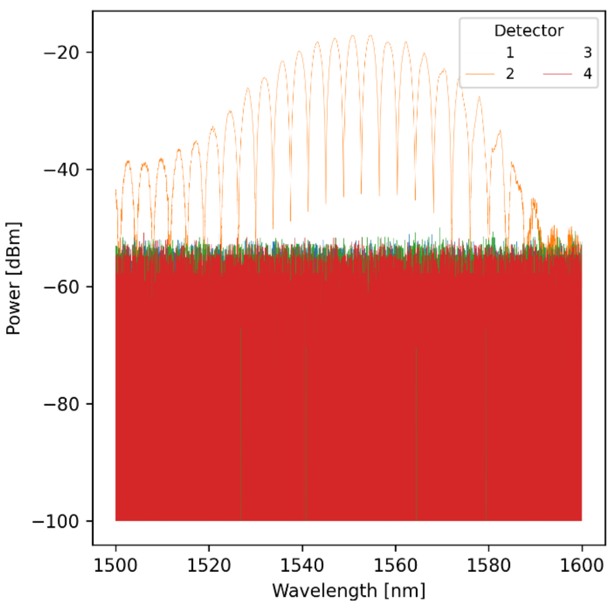

Fig. 9. Measured spectrum of MZI6, DL=150um.

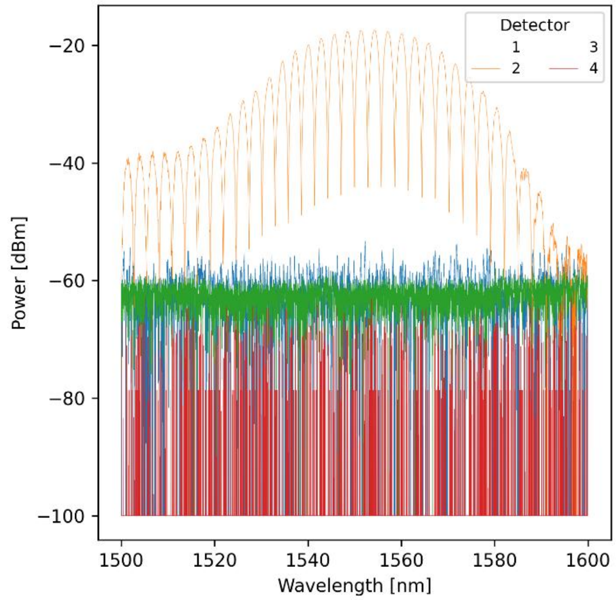

Fig. 10. Measured spectrum of MZI7, DL=200um.

VI. . MEASUREMENTS ANALYSIS

The objective of the analysis is to extract the waveguide parameters given the spectrum of the MZI and the known difference between the MZI arms. The process is done using a python code that was converted from a Matlab code given in [6] with the help of ChatGPT. The analysis process is divided to several steps that will be shortly reviewed next.

The first step of the analysis is to calibrate out the finite bandwidth response of the grading couplers and as a result obtain flat oscillatory spectrum of pure MZI. The method used is by fitting the spectrum to a 4^{th} order polynomial which captures to slow spectrum change with wavelength. This response is subtracted from the spectrum leaving a flat spectrum. Figure 11 shows the result of this process as done over MZI6 spectrum

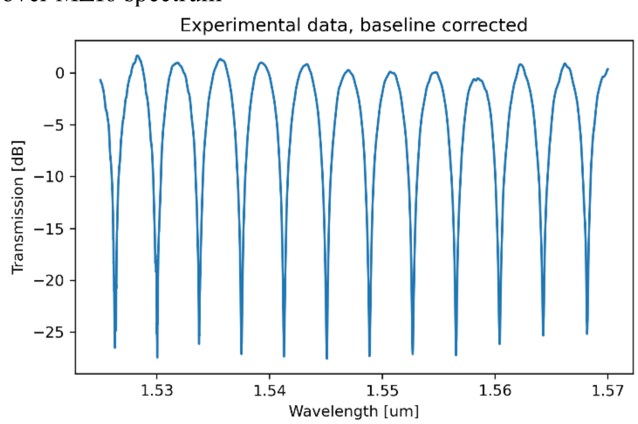

Fig. 11. Spectrum of MZI6 after baseline correction using 4^{th} order polynomial.

Next the following theoretical MZI spectrum function is fitted to the flattened spectrum data using the least square optimization employing the Scipy curve_fit function.

$$F = 10\log_{10}\left(\frac{1}{4}\left|1+\exp\left[-\iota\frac{2\pi n_{eff}}{\lambda}\Delta L-\frac{\alpha}{2}\Delta L\right]\right|^2\right)+b \qquad (6)$$

Where the given parameters are the MZI path difference ΔL and wavelength. The model parameters to be found by optimization are the effective index dispersion relations which is approximated as second order polynomial with three unknown parameters n1, n2 and n3 as in equation (3), and the waveguide propagation loss α the excess loss parameter b. As written in equation (4), the dispersion parameters are directly related to the waveguide effective index, group index and dispassion.

As curve fitting optimization is sensitive to the initial parameters, the code calculates good guess for initial parameters based on the spectrum data as follows. First the parameters n3, α and b are taken as 0, 1e-3 and 0 respectively and are not extracted from the data. The remaining two parameters n1 and n2 which are directly related to the effective and group indices are extracted from the data as follows.

The spectrum free spectral range is estimated by performing autocorrelation to calculate its periodicity. From the estimated free spectral range an initial guess for the group index can be calculated using equation (2). Parameters n1, the effective index has the effect of shifting the spectrum left to right. Plotting the model using an initial guess of n1=2.4, resulting in a shift of the spectrum compared to the data. This shift can be calculated by cross correlation and thus correct the initial guess of n1.

Figure 12 shows the result of the autocorrelation as function of sample lag performed on the spectrum of MZI6, giving a free spectral range of 3.8nm by finding the second peak indicating the shift by one period.

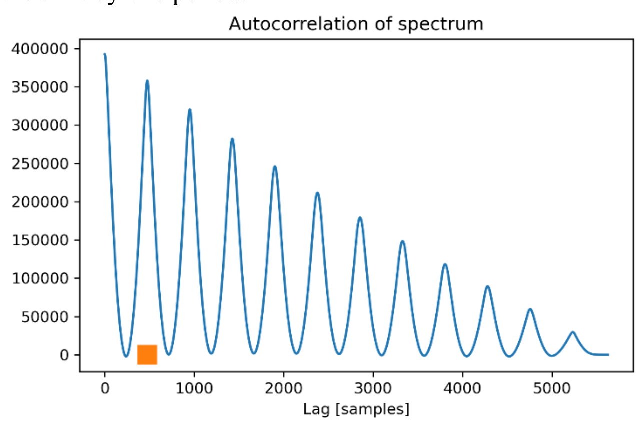

Fig. 12. Autocorrelation of the spectrum of MZI6

Figure 13 shows the model spectrum using the extracted initial guess compared to the measured spectrum of MZI6.

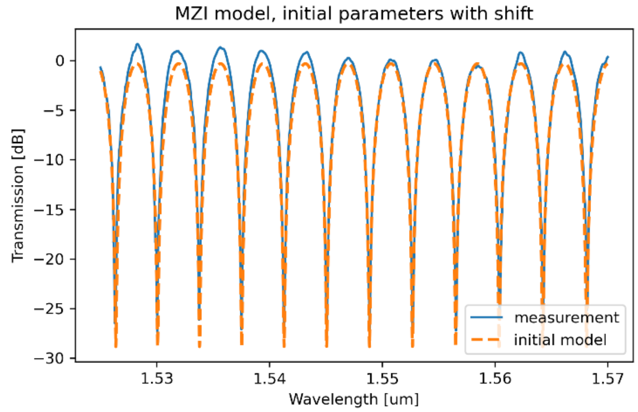

Fig. 13. Model using initial parameters vs measured spectrum of MZI6

After extracting the initial parameters, they are used to performe the least square optimization and arriving to the optimized model as can be seen in figure 14

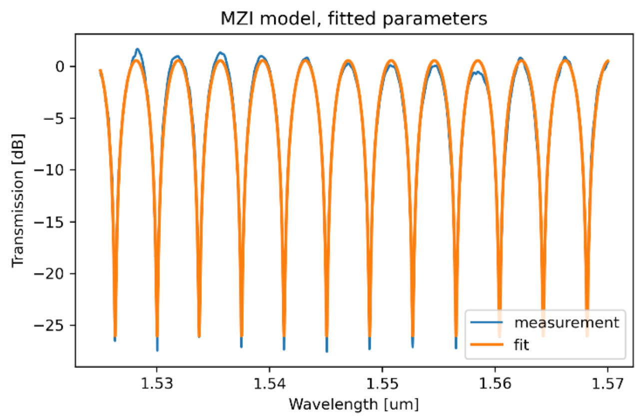

Fig. 14. Fitted model of MZI6

From the fitted model the, using the effective index polynomial the group index as function of wavelength can be calculated from

$$n_g = n_{eff}(\lambda) - \lambda \frac{dn_{eff}(\lambda)}{d\lambda} \tag{7}$$

Where the derivative is calculated numerically. The group index at the center of the frequency can be calculated from equation (4).

The calculated group index as function of wavelength for MZI6 can be seen in figure 15. Where the group index at the center frequency is calculated to be 4.1923.

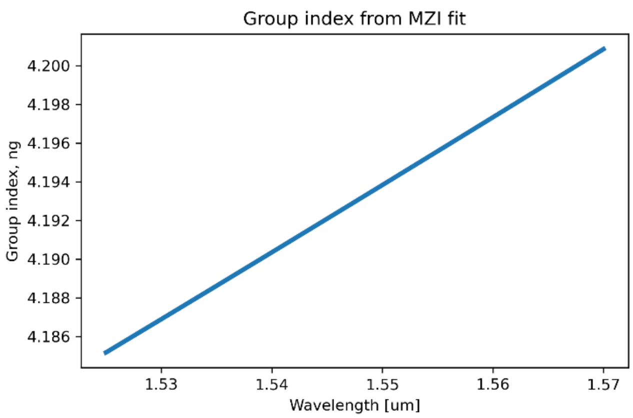

Fig. 15. Extracted group index as function of wavelength from MZI6.

The above process is repeated for all four MZIs (MZI4, MZI5, MZI6 and MZI7). The next table summarize the extracted free spectral range and group index at the center frequencies for each of the discussed MZIs.

TABLE V
EXTRACTED GROUP INDEX AND FSR FROM EACH MZI MEASURED SPECTRUM

MZI	dl [mm]	FSR [nm]	ng
4	75	7.6	4.1952
5	100	5.7	4.1923
6	150	3.8	4.193
7	200	2.848	4.1952

Comparing the measurement results to the corner analysis in tables II and III, we can see that both the FSR and group index extracted from measurements are withing the corner analysis limits simulated due to production variance of waveguide dimensions.

VII CONCLUSIONS

Several MZIs with different optical path-length differences were designed, laid out, fabricated, and measured in a silicon photonics process targeting operation at 1550 nm. A corner analysis was performed for each MZI to estimate process-induced variability in the waveguide group index and the corresponding FSR. The measured transmission spectra were used to extract the FSR and group index of each device. The experimental results show good agreement with the simulated values and fall within the variability predicted by the corner analysis, validating both the design methodology and the process-variation model.

VIII ACKNOWLEDGMENTS

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APPENDIX

The important formulas derivation used in this report are summarized here, they are reproduction of information available at [2] and [6].

A. MZI Transfer Function

In a MZI topology the input light is split using a y-spitter, then each half of the light pulse propagates over a different length waveguide, the two waveguides are then combined using a second y-combiner yielding the output. For input intensity and field of Ii and E_i and neglecting loss the fields at the two inputs of the second combiner arms after each propagated over different length are

$$E_{01} = \frac{E_i}{\sqrt{2}} e^{-i\beta L}$$

And

$$E_{02} = \frac{E_i}{\sqrt{2}} e^{-i\beta(L+\Delta L)}$$

Where β is the propagation constant related to the index of refraction n with

$$\beta = \frac{2\pi n}{\lambda}$$

When both fields are combined, the output intensity becomes

$$I_o = \frac{I_i}{4}\left|e^{-i\beta L} + e^{-i\beta(L+\Delta L)}\right|^2 = \frac{I_i}{2}(1+\cos(\beta\Delta L))$$

The distance between the peaks of this function is called the free spectral range.

B. The Free Spectral Range of Imbalanced MZI

Assuming two adjacent peaks of the MZI transfer function are located at wavelengths λ_{m+1} and λ_m where the distance between them is defined as the FSR

$$FSR = \Delta\lambda = \lambda_{m+1} - \lambda_m$$

Assuming the propagation constant for each of the above adjacent wavelength are β_{m+1} and β_m than the difference between the two propagation constants is defined as

$$\Delta\beta = \beta_m - \beta_{m+1}$$

From the MZI transfer function the phase difference between adjacent peak must be 2π requiring

$$\Delta\beta\Delta L = 2\pi$$

Thus

$$\Delta\beta = \frac{2\pi}{\Delta L}$$

Assuming the distance between the MZI arms is much smaller than the difference between the arms we can approximate the relation between Δβ and Δλ using a derivative

$$\frac{\Delta\beta}{\Delta\lambda} \approx -\frac{d\beta}{d\lambda} = -\frac{d}{d\lambda}\left(\frac{2\pi n}{\lambda}\right) = \frac{2\pi}{\lambda^2}\left(n - \frac{dn}{d\lambda}\lambda\right) = \frac{2\pi}{\lambda^2} n_g$$

Where n_g is the group index.
Thus we can calculate the FSR as follows

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$$\Delta\lambda = \frac{\Delta\beta}{\frac{2\pi}{\lambda^2} n_g} = \frac{\frac{2\pi}{\Delta L}}{\frac{2\pi}{\lambda^2} n_g} = \frac{\lambda^2}{\mathrm{n_g}\Delta L}$$

REFERENCES

[1] R. J. Bojko, J. Li, L. He, T. Baehr-Jones, M. Hochberg, and Y. Aida, "Electron beam lithography writing strategies for low loss, high confinement silicon optical waveguides," J. Vacuum Sci. Technol. B 29, 06F309 (2011)

[2] Lukas Chrostowski, Michael Hochberg, chapter 12 in "Silicon Photonics Design: From Devices to Systems", Cambridge University Press, 2015

[3] http://siepic.ubc.ca/probestation, using Python code developed by Michael Caverley.

[4] Yun Wang, Xu Wang, Jonas Flueckiger, Han Yun, Wei Shi, Richard Bojko, Nicolas A. F. Jaeger, Lukas Chrostowski, "Focusing sub-wavelength grating couplers with low back reflections for rapid prototyping of silicon photonic circuits", Optics Express Vol. 22, Issue 17, pp. 20652-20662 (2014) doi: 10.1364/OE.22.020652

[5] www.plcconnections.com, PLC Connections, Columbus OH, USA.

[6] http://mapleleafphotonics.com, Maple Leaf Photonics, Seattle WA, USA.

[7] L. Chrostowski, "Silicon Photonics Design, Fabrication and Data Analysis," UBCx, edX. Accessed: May 4, 2026. [Online]. Available: https://www.edx.org/learn/engineering/university-of-british-columbia-silicon-photonics-design-fabrication-and-data-ana